Bellman equations for a stochastic policy $\pi(a|s)$

Condition on the first action, then on the next state:

$$V^\pi(s) = E[r_{t+1}|s_t = s] + \gamma\, E[G_{t+1}|s_t = s]$$
$$= \sum_a \pi(a|s) \sum_{s'} p(s'|s,a)\, R(s,a,s')$$
$$+ \gamma \sum_a \pi(a|s) \sum_{s'} p(s'|s,a)\, E[G_{t+1} \,|\, s_t = s, a_t = a, s_{t+1} = s']$$
$$= \sum_a \pi(a|s)\, Q^\pi(s,a),$$
$$Q^\pi(s,a) = \sum_{s'} p(s'|s,a)\big[R(s,a,s') + \gamma V^\pi(s')\big].$$

Substitute one into the other ($R = R(s,a,s')$):

$$Q^\pi(s,a) = \textstyle\sum_{s'} p(s'|s,a)\big[R + \gamma \sum_{a'} \pi(a'|s')\, Q^\pi(s',a')\big]$$

Optimal values: $V^*(s) = \max_a Q^*(s,a)$, so

$$Q^*(s,a) = \textstyle\sum_{s'} p(s'|s,a)\big[R + \gamma \max_{a'} Q^*(s',a')\big]$$

Each equation says: $Q(s,a)$ is the mean of a random variable built from one transition $(s,a) \to (r,s')$:

$$Q(s,a) = \mathbb{E}\big[\, Y \mid s_t = s,\ a_t = a \,\big].$$

	target Y	mean
SARSA	$r + \gamma\, Q(s',a')$	Q^π
	$s' \sim p(\cdot\|s,a)$, $a' \sim \pi(\cdot\|s')$ sampled	
Exp. SARSA	$r + \gamma \sum_b \pi(b\|s')\, Q(s',b)$	Q^π
	only s' sampled; sum over a' exact	
Q-learning	$r + \gamma \max_b Q(s',b)$	Q^*
	only s' sampled; a' not needed	

SARSA and Expected SARSA: same mean, smaller variance for Expected SARSA (no noise from a').

From the expectation to an update: mini-batch, then Robbins–Monro

1. If Q on the right were known: a mini-batch Monte Carlo estimate. Draw B transitions from (s, a) and average:

$$Q(s,a) \approx \bar{Y} = \frac{1}{B}\sum_{i=1}^{B} Y_i \qquad (\text{LLN as } B \to \infty).$$

2. But Q inside Y is the unknown. Use the old table Q_k in the targets:

$$Q_{k+1}(s,a) = \bar{Y}_k = (TQ_k)(s,a) + \text{noise},$$

T: the Bellman operator. With $B = \infty$ this is value iteration (simple iterations), convergent since T is a γ-contraction.

3. Robbins–Monro: do not wait for a large batch; take a step towards it:

$$Q_{k+1}(s,a) = Q_k(s,a) + \alpha_k\big(\bar{Y}_k - Q_k(s,a)\big)$$

Why it converges. We look for the root of $g(Q) = TQ - Q = 0$, observed only with noise:

$$\bar{Y}_k - Q_k = g(Q_k) + \xi_k, \qquad \mathbb{E}[\xi_k \mid Q_k] = 0.$$

Robbins–Monro conditions, for each pair (s, a) with k counting its own updates:

$$\sum_k \alpha_k = \infty, \quad \sum_k \alpha_k^2 < \infty, \quad \operatorname{Var} \xi_k \le \sigma^2,$$

and every (s, a) updated infinitely often $\Rightarrow Q_k \to$ the fixed point (Q^π or Q^*).

algorithm	batch B	step α
value iteration	∞ (model)	1
SARSA, Exp. SARSA, QL	1	const, $n^{-0.7}$
Q-learning sweep	1 per pair	$n^{-0.7}$
DQN	64 from replay	Adam step